Contents

1 Part 1

Your friend has set up a website for her new business. So far $N = 115$ potential customers has visited her site, but only $y = 4$ customers have completed a purchase. To plan her future investments, she asks you for help to compute the probability that at least one of the next $N^* = 20$ customers will make a purchase. You decide to model the problem using the beta-binomial model with a uniform prior distribution on the probability of making a purchase $\theta \in [0, 1]$:

$$\theta \sim \text{Beta}(a_0, b_0) \tag{1.1}$$

$$y|\theta \sim \text{Binomial}(N, \theta) \tag{1.2}$$

where $a_0 = b_0 = 1$

1.1 Task 1.1 | Compute the prior mean of θ and provide a 95%-credibility interval for the prior.

The prior of θ follows a beta distribution given in Equation (1.3):

$$\text{Prior:} \quad p(\theta) = \text{Beta}(\theta|, a_0, b_0) \tag{1.3}$$

With the prior mean given as:

$$\mathbb{E}[\theta] = \int_0^1 \theta \; p(\theta|a_0, b_0) \; d\theta = \frac{a_0}{a_0 + b_0} \tag{1.4}$$

Inserting the known values in Equation (1.4), results in:

$$\mathbb{E}[\theta] = \frac{1}{1+1} = \frac{1}{2} = 0.5 \tag{1.5}$$

Since our a_0, b_0 parameters are both 1, we can easily identify the 95 %-credibility interval to be:

$$[CI_{\text{lower}}, CI_{\text{upper}}] = \left[\frac{\alpha}{2}, 1 - \frac{\alpha}{2}\right] \tag{1.6}$$

Where $\alpha = 1 - 0.95 = 0.05$, giving us the credibility interval $[0.025, 0.975]$

1.2 Task 1.2 | Compute the posterior mean of θ and provide a 95%-credibility interval for the posterior.

Before computing the posterior, we first have to calculate the posterior parameters a and b, which are given by:

$$a = a_0 + y, \qquad b = b_0 + N - y \tag{1.7}$$

Inserting our given values, gives us:

$$a = 1 + 4 = 5, \qquad b = 1 + 115 - 4 = 112 \tag{1.8}$$

The posterior mean is then given by the formula:

$$\mathbb{E}[\theta|\mathcal{D}] = \frac{a}{a+b} \tag{1.9}$$

Using Equation (1.9), we calculate the posterior mean to be:

$$\mathbb{E}[\theta|\mathcal{D}] = \frac{a}{a+b} = \frac{5}{5+112} = \frac{5}{117} \approx 0.043 \tag{1.10}$$

To calculate the credibility interval for the posterior, we use the inverse CDF (F^{-1}) to get our intervals:

$$CI = \left[F^{-1}\left(\frac{\alpha}{2}\right), F^{-1}\left(1-\frac{\alpha}{2}\right)\right], \quad \text{where} \quad \alpha = 1 - 0.95. \tag{1.11}$$

This was calculated using the `beta_dist.interval` function in python, and returned the interval:

$$CI = [0.01, 0.09] \tag{1.12}$$

1.3 Task 1.3 | Compute and plot the posterior predictive distribution for y^* given $y = 4$.

Via the lecture from week 2, it is known that the posterior predictive distribution $p(y^* = k|20)$ is given as:

$$p(y^* = k|y) = \binom{N^*}{k} \frac{B(\alpha + k, \beta + N^* - k)}{B(\alpha, \beta)} \tag{1.13}$$

We use the values states in the description, where $y = 4$, $N^* = 20$, $\alpha = 5$ and $\beta = 112$.

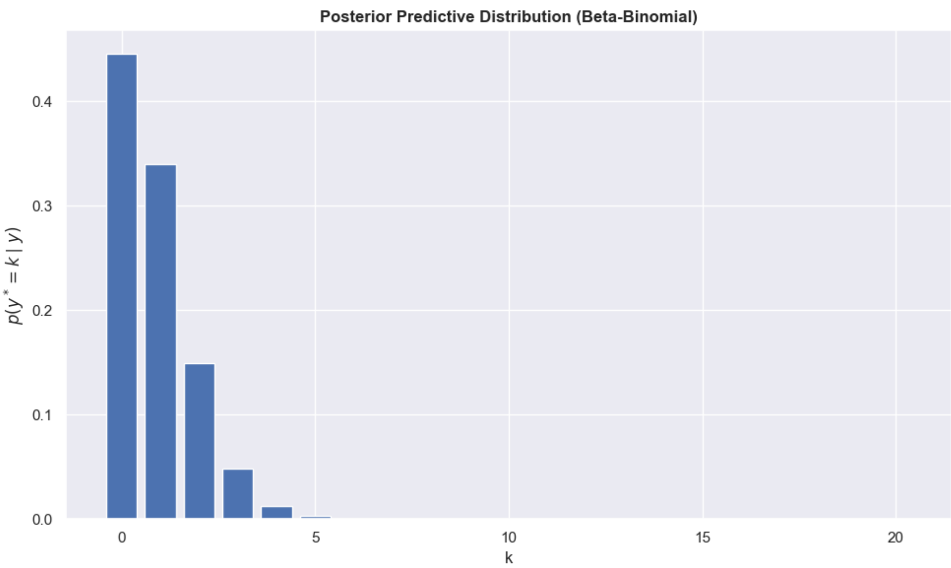

Figure 1.1: Posterior predictive distribution of 4 completed purchases within 20 views.

1.4 Task 1.4 | What is the posterior predictive probability that at least one of the next $N^* = 20$ customers will make a purchase?

We now want to calculate $p(y^* \geq 1|y = 4)$, which is equal to:

$$p(y^* \geq 1|y = 4) = 1 - p(y^* = 0|y = 4) = 0.55 \tag{1.14}$$

The posterior predictive probability that at least one customer out of the next 20 will make a purchase is, therefore, $p(y^* \geq 1|y = 4) = 0.55$.

1.5 Task 1.5 | Compute mean and variance of the posterior predictive distribution for y^*.

We use the following formula to compute the mean of the posterior predictive distribution:

$$\mathbb{E}[k] = \sum_{k}^{N^*} k \cdot p(k) \approx 0.85 \tag{1.15}$$

then, using the mean value, we can further compute the variance using the following formula:

$$\mathbb{V}[k] = \sum_{k}^{N^*} k^2 \cdot p(k) - (\mathbb{E}(k))^2 \approx 0.95 \tag{1.16}$$

Therefore, the mean is $\mathbb{E}[k] = 0.85$ with variance $\mathbb{V}[k] = 0.95$.

2 Part 2

Let $\boldsymbol{z_1}$, $\boldsymbol{z_2} \in \mathbb{R}^2$, and $y \in \mathbb{R}$ be random variables and consider the following Gaussian system

$$\boldsymbol{z_1} \sim \mathcal{N}(\mathbf{0}, v\boldsymbol{I}) \tag{2.1}$$

$$\boldsymbol{z_2}|\boldsymbol{z_1} \sim \mathcal{N}(\boldsymbol{z_1}, v\boldsymbol{I}) \tag{2.2}$$

$$y|\boldsymbol{z_2} \sim \mathcal{N}(\boldsymbol{a}^T\boldsymbol{z_2}, \sigma^2) \tag{2.3}$$

where $\boldsymbol{a} \in \mathbb{R}^2$ is a constant. The joint distribution of $(\boldsymbol{z_1}, \boldsymbol{z_2}, y)$ is given by:

$$p(y, \boldsymbol{z_1}, \boldsymbol{z_2}) = p(y|\boldsymbol{z_2})p(\boldsymbol{z_2}|\boldsymbol{z_1})p(\boldsymbol{z_1}) \tag{2.4}$$

2.1 Task 2.1 | Determine the distribution $p(y)$.

$p(y)$ can be obtained from the joint distribution via the sum rule, hence, we write:

$$p(y) = \int\int p(y, \boldsymbol{z_1}, \boldsymbol{z_2})d\boldsymbol{z_1}d\boldsymbol{z_2} \tag{2.5}$$

We insert the 2.4 into 2.5 and get:

$$p(y) = \int\int p(y|\boldsymbol{z_2})p(\boldsymbol{z_2}|\boldsymbol{z_1})p(\boldsymbol{z_1})d\boldsymbol{z_1}d\boldsymbol{z_2} \tag{2.6}$$

We turn to Gaussian properties and rules in order to solve the integral. With the information given in the task description, we see that $\boldsymbol{z_1} \sim \mathcal{N}(\mathbf{0}, v\boldsymbol{I})$ and $\boldsymbol{z_2}|\boldsymbol{z_1} \sim \mathcal{N}(\boldsymbol{z_1}, v\boldsymbol{I})$, indicating the following (done by using the rules on page 7 in week 3 slides):

$$\boldsymbol{z_2} = \boldsymbol{z_1} + \epsilon_2 \tag{2.7}$$

With $\epsilon_2 \sim \mathcal{N}(0, v\boldsymbol{I})$, and with this information we use the rule that the sum of independent Gaussians is also Gaussian:

$$z_2 \sim \mathcal{N}(0, 2v\boldsymbol{I}) \tag{2.8}$$

Furthermore, we were given $y|\boldsymbol{z_2} \sim \mathcal{N}(\boldsymbol{a}^T\boldsymbol{z_2}, \sigma^2)$, where we can write:

$$y = \boldsymbol{a}^T\boldsymbol{z_2} + \epsilon_y \tag{2.9}$$

Where $\epsilon_y \sim \mathcal{N}(0, \sigma^2)$. Then if we take the following properties from the slides:

$$x \sim \mathcal{N}(m_x, V_x) \tag{2.10}$$

$$a + Bx \sim \mathcal{N}(a + Bm_x, BV_xB^T) \tag{2.11}$$

In our case, $B = a^T$ and $x = \boldsymbol{z_2}$ and $b = 0$, and from 2.8, the mean is then:

$$\mathbb{E}[y] = \boldsymbol{a}^T\mathbb{E}[\boldsymbol{z_2}] = 0 \tag{2.12}$$

Variance is:

$$V[y] = V[a^T\boldsymbol{z_2}] + V[\epsilon_y] = a^T(2vI)a + \sigma^2 \tag{2.13}$$

Therefore, we get:

$$y \sim \mathcal{N}(0, a^T(2vI)a + \sigma^2) \tag{2.14}$$

Thus, the marginal distribution of y becomes:

$$p(y) = \mathcal{N}\left(0,\, \boldsymbol{a}^T(2v\boldsymbol{I})\boldsymbol{a} + \sigma^2\right). \tag{2.15}$$

2.2 Task 2.2 | Determine the distribution $p(y, \boldsymbol{z_2}|\boldsymbol{z_1})$.

We want to determine $p(y, \boldsymbol{z_2}|\boldsymbol{z_1})$ and using the product rule gives (see page 9, slides from week 2):

$$p(y, \boldsymbol{z_2}|\boldsymbol{z_1}) = p(y|\boldsymbol{z_2})p(\boldsymbol{z_2}|\boldsymbol{z_1}) \tag{2.16}$$

Now we just insert the Gaussian forms of the model given in the task description:

$$\boldsymbol{z_2}|\boldsymbol{z_1} \sim \mathcal{N}(\boldsymbol{z_1}, v\boldsymbol{I}) \tag{2.17}$$

$$y|\boldsymbol{z_1} \sim \boldsymbol{a}^T\boldsymbol{z_2}, \sigma^{\in} \tag{2.18}$$

Inserting them into 2.16 gives us:

$$p(y, \boldsymbol{z_2}|\boldsymbol{z_1}) = \mathcal{N}(y \mid \boldsymbol{a}^T\boldsymbol{z_2}, \sigma^2)\mathcal{N}(\boldsymbol{z_2} \mid \boldsymbol{z_1}, v\boldsymbol{I}) \tag{2.19}$$

Which concludes this task.

2.3 Task 2.3 | Determine the distribution $p(y|\boldsymbol{z_1})$.

We have to use the sum rule (again from week 2), and we obtain:

$$p(y|\boldsymbol{z_1}) = \int p(y, \boldsymbol{z_2}|\boldsymbol{z_1})d\boldsymbol{z_2} \tag{2.20}$$

The result from the previous task 2.2 is inserted in 2.20 to give:

$$p(y|\boldsymbol{z_1}) = \int \mathcal{N}(y|\boldsymbol{a}^T\boldsymbol{z_2}, \sigma^2)\mathcal{N}(\boldsymbol{z_2}|\boldsymbol{z_1}, v\boldsymbol{I})d\boldsymbol{z_2}. \tag{2.21}$$

Using the linear Gaussian model $\boldsymbol{z_2} = \boldsymbol{z_1} + \epsilon_2$ and $y = \boldsymbol{a}^T\boldsymbol{z_2} + \epsilon_y$, we obtain:

$$y = \boldsymbol{a}^T\boldsymbol{z_1} + \boldsymbol{a}^T\epsilon_2 + \epsilon_y \tag{2.22}$$

With $\epsilon_2 \sim \mathcal{N}(0, v\boldsymbol{I})$ and $\epsilon_y \sim \mathcal{N}(0, \sigma^2)$, where:

$$\boldsymbol{a}^T\epsilon_2 \sim \mathcal{N}(0, v\boldsymbol{a}^T\boldsymbol{a}) \tag{2.23}$$

Finally, inserting everything yields:

$$p(y|\boldsymbol{z_1}) = \mathcal{N}(\boldsymbol{a}^T\boldsymbol{z_1},\ v\boldsymbol{a}^T\boldsymbol{a} + \sigma^2). \tag{2.24}$$

We have hereby determined the distribution of $p(y|\boldsymbol{z_1})$.

2.4 Task 2.4 | Determine the distribution $p(\boldsymbol{z_1}|y)$.

We start by using Bayes' rule:

$$p(\boldsymbol{z_1}|y) = \frac{p(y|\boldsymbol{z_1})p(\boldsymbol{z_1})}{p(y)} \tag{2.25}$$

From the previous tasks we have:

$$p(\boldsymbol{z_1}) = \mathcal{N}(\boldsymbol{0}, v\boldsymbol{I}) \tag{2.26}$$

$$p(y|\boldsymbol{z_1}) = \mathcal{N}(\boldsymbol{a}^T\boldsymbol{z_1}, v\boldsymbol{a}^T\boldsymbol{a} + \sigma^2). \tag{2.27}$$

Now using the Gaussian conditioning formulas for linear Gaussian systems (Murphy section 3.3), the posterior precision is:

$$\boldsymbol{\Sigma}^{-1} = \boldsymbol{\Sigma}_z^{-1} + \boldsymbol{W}^T\boldsymbol{\Sigma}_y^{-1}\boldsymbol{W} \tag{2.28}$$

From the model we identify $\boldsymbol{\Sigma}_z = v\boldsymbol{I}$, $\boldsymbol{W} = \boldsymbol{a}^T$, $\boldsymbol{\Sigma}_y = v\boldsymbol{a}^T\boldsymbol{a} + \sigma^2$. Substituting these into the expression gives:

$$\boldsymbol{\Sigma}^{-1} = \frac{1}{v}\boldsymbol{I} + \frac{1}{v\boldsymbol{a}^T\boldsymbol{a} + \sigma^2}\boldsymbol{a}\boldsymbol{a}^T \tag{2.29}$$

Therefore:

$$\boldsymbol{\Sigma} = \left(\frac{1}{v}\boldsymbol{I} + \frac{1}{v\boldsymbol{a}^T\boldsymbol{a} + \sigma^2}\boldsymbol{a}\boldsymbol{a}^T\right)^{-1} \tag{2.30}$$

Similarly, the posterior mean is:

$$\boldsymbol{\mu} = \boldsymbol{\Sigma}\boldsymbol{W}^T\boldsymbol{\Sigma}_y^{-1}y = \boldsymbol{\Sigma}\boldsymbol{a}\frac{1}{v\boldsymbol{a}^T\boldsymbol{a} + \sigma^2}y \tag{2.31}$$

Thus giving us the final result:

$$p(\boldsymbol{z_1}|y) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \tag{2.32}$$

Which concludes this exercise.

3 Part 3

3.1 Task 3.1 | Determine the joint distribution of $(\boldsymbol{y}, \lambda)$

Using the product rule of probability introduced in week 1, the joint distribution is given by the product of the likelihood and the prior. Since the observations y_i are assumed to be conditionally independent given λ, we have:

$$p(\boldsymbol{y}, \lambda) = p(\boldsymbol{y}|\lambda)p(\lambda) \tag{3.1}$$

The likelihood function for N independent Poisson-distributed variables is:

$$p(\boldsymbol{y}|\lambda) = \prod_{i=1}^{N} \frac{\lambda^{y_i} e^{-\lambda}}{y_i!} = \frac{\lambda^{\sum_{i=1}^{N} y_i} e^{-N\lambda}}{\prod_{i=1}^{N} y_i!} \tag{3.2}$$

The prior distribution for λ is a Gamma distribution:

$$p(\lambda) = \frac{b_0^{a_0}}{\Gamma(a_0)} \lambda^{a_0-1} e^{-b_0\lambda} \tag{3.3}$$

Multiplying these two expressions yields the joint distribution:

$$\begin{aligned} p(\boldsymbol{y}, \lambda) &= \left(\frac{\lambda^{\sum_{i=1}^{N} y_i} e^{-N\lambda}}{\prod_{i=1}^{N} y_i!} \right) \left(\frac{b_0^{a_0}}{\Gamma(a_0)} \lambda^{a_0-1} e^{-b_0\lambda} \right) \\ &= \frac{b_0^{a_0}}{\Gamma(a_0) \prod_{i=1}^{N} y_i!} \lambda^{a_0-1+\sum_{i=1}^{N} y_i} e^{-\lambda(b_0+N)} \end{aligned} \tag{3.4}$$

3.2 Task 3.2 | Show that the functional form of a Gamma distribution is given by $\log p(\lambda|a, b) = (a-1)\log(\lambda) - b\lambda +$ *constant*.

Following the methodology for working with conjugate models (week 2), we can identify the distribution by isolating its functional form. We take the natural logarithm of the Gamma probability density function:

$$\log p(\lambda|a, b) = \log \left(\frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} \right) \tag{3.5}$$

Using logarithm rules, this expands to:

$$\log p(\lambda|a, b) = \log \left(\frac{b^a}{\Gamma(a)} \right) + (a-1)\log(\lambda) - b\lambda \tag{3.6}$$

The term $\log\left(\frac{b^a}{\Gamma(a)}\right)$ consists purely of the constants a and b and is independent of λ. Absorbing this into a constant leaves us with the functional form:

$$\log p(\lambda|a, b) = (a-1)\log(\lambda) - b\lambda + \text{constant} \tag{3.7}$$

3.3 Task 3.3 | Derive the analytical expression for the posterior distribution $p(\lambda|\boldsymbol{y})$ and show that it is a Gamma-distribution.

According to Bayes' theorem from week 1, the posterior distribution is proportional to the likelihood multiplied by the prior:

$$p(\lambda|\boldsymbol{y}) \propto p(\boldsymbol{y}|\lambda)p(\lambda) \tag{3.8}$$

Taking the logarithm on both sides yields:

$$\log p(\lambda|\boldsymbol{y}) = \log p(\boldsymbol{y}|\lambda) + \log p(\lambda) + C \tag{3.9}$$

We insert the respective log-expressions:

$$\log p(\boldsymbol{y}|\lambda) = \sum_{i=1}^{N} (y_i \log(\lambda) - \lambda - \log(y_i!)) = \log(\lambda) \sum_{i=1}^{N} y_i - N\lambda - \sum_{i=1}^{N} \log(y_i!) \tag{3.10}$$

$$\log p(\lambda) = (a_0 - 1) \log(\lambda) - b_0\lambda + \text{constant} \tag{3.11}$$

Summing these terms and absorbing everything that does not depend on λ into a new constant, we get:

$$\log p(\lambda|\boldsymbol{y}) = \left(a_0 - 1 + \sum_{i=1}^{N} y_i \right) \log(\lambda) - (b_0 + N)\lambda + \text{constant} \tag{3.12}$$

This perfectly matches the functional form of a Gamma distribution derived in Task 3.2. By matching the terms, we identify the updated posterior parameters:

$$a_N = a_0 + \sum_{i=1}^{N} y_i \tag{3.13}$$

$$b_N = b_0 + N \tag{3.14}$$

This analytical derivation confirms that the Poisson-Gamma model is a conjugate system (as defined in week 2) and that the posterior distribution is indeed a Gamma distribution: $p(\lambda|\boldsymbol{y}) = \text{Gamma}(a_N, b_N)$.

3.4 Task 3.4 | Determine the posterior distribution for λ given the data above and report the mean.

Given the data $\boldsymbol{y} = [7, 4, 8, 11, 12]$ and $N = 5$. The sum of the observations is:

$$\sum_{i=1}^{5} y_i = 7 + 4 + 8 + 11 + 12 = 42 \tag{3.15}$$

The prior parameters are given as $a_0 = 1$ and $b_0 = 0.1$. Inserting this into our posterior parameters gives:

$$a_N = 1 + 42 = 43 \tag{3.16}$$

$$b_N = 0.1 + 5 = 5.1 \tag{3.17}$$

The posterior distribution for λ is therefore $\text{Gamma}(43, 5.1)$. The expected value (mean) of the posterior is calculated as:

$$\mathbb{E}[\lambda|\boldsymbol{y}] = \frac{a_N}{b_N} = \frac{43}{5.1} \approx 8.431 \tag{3.18}$$

3.5 Task 3.5 | Plot $p(\lambda)$ and $p(\lambda|\boldsymbol{y})$ for $\lambda \in [0, 30]$.

The plot was generated using Python and *scipy.stats.gamma*, comparing the prior and posterior distributions.

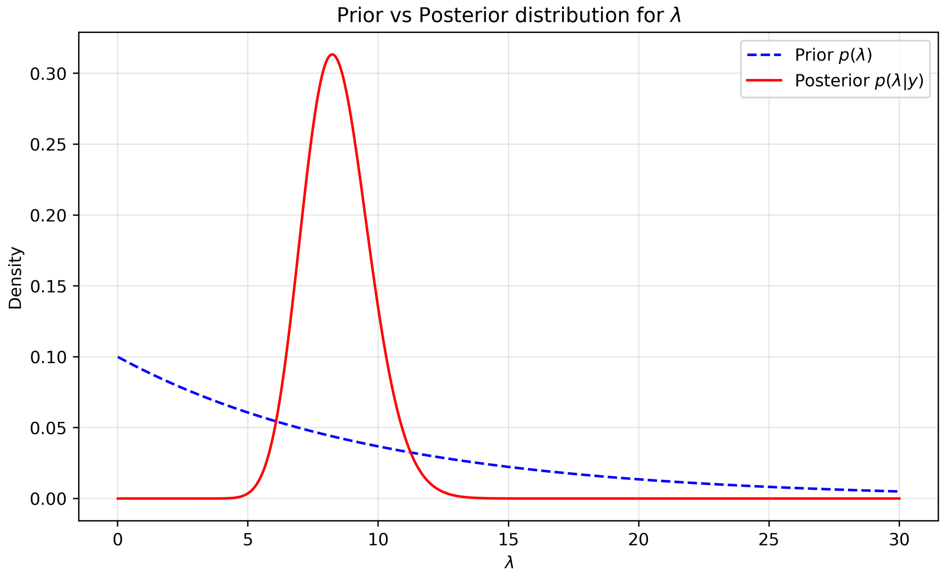

Figure 3.1: Prior vs Posterior distribution for λ.

As seen in Figure 3.1, the prior distribution is relatively flat and diffuse, reflecting a lack of strong prior knowledge regarding the rate parameter λ. After incorporating the observed data, the posterior distribution becomes significantly narrower and is peaked around its calculated mean of roughly 8.43. This visualization demonstrates how the evidence provided by the data heavily updates the initial belief, substantially reducing the uncertainty about the true value of λ.